The nn-bounded operators are not closed under nn-convergence

A counterexample to a question in arXiv:math/0004049

Source Question

Troitsky defines nn-convergence for operators on a topological vector space and asks whether the class of nn-bounded operators is closed relative to that convergence [1].

> The class of all nn-bounded operators will be usually equipped with the topology of ***nn-convergence***, defined as follows. We will call a collection $\mathcal{G}$ of operators ***uniformly nn-bounded*** if there exists a base $\mathcal{N}_0$ of zero neighborhoods such that for every $U \in \mathcal{N}_0$ there exists a positive real β such that $S(U) \subseteq \beta U$ for each $S \in \mathcal{G}$. We say that a sequence (S_n) ***nn-converges*** to zero if there is a base $\mathcal{N}_0$ of zero neighborhoods such that for every $U \in \mathcal{N}_0$ and every $\varepsilon > 0$ we have $S_n(U) \subseteq \varepsilon U$ for all sufficiently large n.
>
> **Question.** *Is the class of all nn-bounded operators closed relative to nn-convergence?*

Figure 1: The definition of nn-convergence and the displayed question.

The source paper also records that the left shift on $\mathbb{R}^{\mathbb{N}}$, with the topology of coordinatewise convergence, is continuous but not nn-bounded.

> **Example 2.5.** *A continuous but not nn-bounded operator.* Let T be the left shift on the space of all real sequences $\mathbb{R}^{\mathbb{N}}$ with the topology of coordinate-wise convergence, i.e., $T\colon (x_0, x_1, x_2, \dots) \mapsto (x_1, x_2, x_3, \dots)$. Clearly T is continuous. We will show that T is not nn-bounded. Assume that for every zero neighborhood U in some base $\mathcal{N}_0$ there is a positive scalar α such that $T(U) \subseteq \alpha U$. Since the set $\bigl\{x = (x_k) : |x_0| < 1\bigr\}$ is a zero neighborhood, there must be a base neighborhood $U \in \mathcal{N}_0$ such that $U \subseteq \{x : |x_0| < 1\}$. Since $T(U) \subseteq \alpha U$ for some positive α then $T^n(U) \subseteq \alpha^n U$, so that if $x = (x_k) \in U$ then $T^n x \in \alpha^n U$, so that $|x_n| = \bigl|(T^n x)_0\bigr| < \alpha^n$. Hence $U \subseteq \bigl\{x : |x_n| < \alpha^n \text{ for each } n > 0\bigr\}$. But this set is bounded, while the space is not locally bounded, a contradiction.

Figure 2: Example 2.5 in the source paper.

Proof Intuition

The topology of $\mathbb{R}^{\mathbb{N}}$ only tests finitely many coordinates at a time. Finite truncations of the left shift are harmless: each one has finite-dimensional range, hence maps a suitable zero neighborhood into a bounded set. But for any fixed finite coordinate test, a far enough truncation agrees with the full left shift on all tested coordinates. Thus the truncated shifts converge to the full left shift in nn-convergence even though the full shift is not nn-bounded.

Counterexample

Theorem 1. *The class of nn-bounded operators need not be closed relative to nn-convergence.*

Proof. Let $X = \mathbb{R}^{\mathbb{N}}$ with the product topology, where coordinates are indexed by $0, 1, 2, \ldots$. A basic zero neighborhood is a set

$$U(F, \delta) = \{x \in X : |x_j| < \delta_j \text{ for every } j \in F\},$$

where $F \subset \mathbb{N}$ is finite and each $\delta_j > 0$. A subset of X is bounded if and only if it is bounded in each coordinate.

Let $L : X \to X$ be the left shift

$$L(x_0, x_1, x_2, \ldots) = (x_1, x_2, x_3, \ldots).$$

For $N \geq 0$, let P_N be the coordinate projection

$$P_N(y_0, y_1, \ldots) = (y_0, \ldots, y_N, 0, 0, \ldots),$$

and put $S_N = P_N L$. We first show that each S_N is nn-bounded. In fact it is nb-bounded. The zero neighborhood

$$V_N = \{x \in X : |x_j| < 1 \text{ for } 1 \leq j \leq N+1\}$$

is mapped by S_N into the set of all y satisfying $|y_j| < 1$ for $0 \leq j \leq N$ and $y_j = 0$ for $j > N$. This image is coordinatewise bounded, hence bounded in X. Thus S_N is nb-bounded, and by the hierarchy in [1, Proposition 2.1], each S_N is nn-bounded.

Now we prove that $S_N \to L$ in nn-convergence. It is enough to use the standard base $U(F, \delta)$ above. Fix such a $U = U(F, \delta)$ and fix $\varepsilon > 0$. Choose $N \geq \max F$, with the convention that the condition is empty if $F = \varnothing$. For $j \in F$, the j-th coordinate of $(L - S_N)x$ is zero for every $x \in X$, since $P_N L$ keeps all coordinates $0, \ldots, N$ of Lx. Therefore

$$(L - S_N)(U) \subseteq \varepsilon U$$

for all sufficiently large N. Hence $S_N - L$ nn-converges to zero, or equivalently S_N nn-converges to L.

Finally, L is not nn-bounded. For completeness we recall the short argument. Suppose L were nn-bounded. Then there would be a base $\mathcal{N}_0$ of zero neighborhoods such that for every $U \in \mathcal{N}_0$ one has $L(U) \subseteq \alpha U$ for some $\alpha > 0$. Choose $U \in \mathcal{N}_0$ with

$$U \subseteq \{x \in X : |x_0| < 1\}.$$

Then $L^n(U) \subseteq \alpha^n U$, and for every $x \in U$ we get

$$|x_n| = |(L^n x)_0| < \alpha^n \qquad (n \geq 1).$$

Thus U is contained in a coordinatewise bounded set, so U itself is bounded. But $\mathbb{R}^{\mathbb{N}}$ with the product topology has no bounded zero neighborhood: every zero neighborhood leaves some coordinate unrestricted. This contradiction shows that L is not nn-bounded.

The sequence (S_N) therefore consists of nn-bounded operators, nn-converges to L, and has a non-nn-bounded limit. $\square$

Verification and Scope

No computational verification is needed. The only structural facts used are the coordinatewise description of bounded sets in the product topology and the nb-bounded implies nn-bounded implication from the source paper. The result answers the displayed closure question negatively.

Novelty Check

The run indexes were searched for `0004049`, `Spectral Radii`, `nn-bounded`, `nn-convergence`, `left shift`, `product topology`, and `Troitsky`; no prior local packet or attempt was found. Web searches for the exact displayed question and close variants involving `nn-bounded`, `nn-convergence`, and `left shift` did not reveal a separate known answer. This is a bounded novelty check, not an exhaustive literature survey.

References

[1] Vladimir G. Troitsky, *Spectral Radii of Bounded Operators on Topological Vector Spaces*, arXiv:math/0004049, 2000. `https://arxiv.org/abs/math/0004049`

Human Review: nn-Bounded Operators and nn-Convergence

Original automatic proof: `solution_packet.pdf`
Checked date: 15 June 2026
Status: Rejected as literature already answered.
Verdict: This packet should not be counted as a novel counterexample. The question was answered in later literature, specifically in Theorem 2.2 of Zabeti's paper *Topological Algebras of Bounded Operators on Topological Vector Spaces*.

Human Comments

The automatic packet addresses Troitsky's question asking whether the class of nn-bounded operators is closed under nn-convergence. The proposed construction uses finite truncations of the left shift on $\mathbb{R}^{\mathbb{N}}$, with the product topology, converging in nn-convergence to the full left shift, which is not nn-bounded.

The mathematical construction is valid, but it should be excluded from the survey as a new discovery, because the same phenomenon is already treated in the literature:

Zabeti, *Topological Algebras of Bounded Operators on Topological Vector Spaces*,

Journal of Advanced Research in Pure Mathematics 3(1) (2011), 22–26.

The DOI metadata for this paper is

`10.5373/jarpm.438.052210`.

More specifically, Theorem 2.2 of that paper states that, for a locally pseudo-convex topological vector space X, the operations in $B_{nn}(X)$ are continuous for the topology of nn-convergence, and that $B_{nn}(X)$ fails to be closed in this topology. The proof gives the same kind of example: take $X = \mathbb{C}^{\mathbb{N}}$ with the topology of coordinatewise convergence, let S_n be the left shift followed by projection onto the first n coordinates, and observe that S_n nn-converges to the full left shift S, while S is not nn-bounded, as in Troitsky's Example 2.5.

Thus this packet is best classified as *already answered by later work*, rather than as a new counterexample produced by the pipeline. It appears to be a rediscovery of the example in Zabeti's Theorem 2.2. I would therefore reject it for inclusion among novel or potentially novel mathematical outputs.

Review Outcome

Rejected. The reason is not that the construction is incorrect, but that the question was already answered in the later literature cited above. This should be moved conceptually to the literature-already-answered category if the repository classification is updated later.